



Guided Waves in Composite Overwrapped Pressure Vessels and Considerations for Sensor Placement Toward Structural Health Monitoring—An Experimental Study

Jan Heimann¹

Bundesanstalt für Materialforschung
und -prüfung (BAM),
Unter den Eichen 87,
Berlin 12205, Germany
e-mail: jan.heimann@bam.de

Samir Mustapha

Laboratory of Smart Structures and Structural
Integrity (SSSI), Mechanical Engineering,
American University of Beirut,
Beirut 1107 2020, Lebanon
e-mail: sm154@aub.edu.lb

Bengisu Yilmaz

Bundesanstalt für Materialforschung
und -prüfung (BAM),
Unter den Eichen 87,
Berlin 12205, Germany
e-mail: bengisu.yilmaz@bam.de

Jens Prager

Bundesanstalt für Materialforschung
und -prüfung (BAM),
Unter den Eichen 87,
Berlin 12205, Germany
e-mail: jens.prager@bam.de

The utilization of composite overwrapped pressure vessels (COPVs) to store hydrogen, especially at high pressures, is gaining more popularity due to their lightweight design and high storage density, offering significant economic advantages. However, the presence of material defects or fatigue can lead to critical failures, requiring an innovative and robust approach to ensure safe operation and system integrity. Developing a continuous structural health monitoring (SHM) system for COPVs can provide comprehensive real-time information about their condition, facilitating a shift away from periodic inspections. This study scrutinizes the behavior of guided waves (GWs) within COPVs to design a sensor network for damage detection and localization. First, the dispersive and multimodal propagation behavior of GWs is experimentally investigated. Subsequently, important parameters for the network design are derived and finally a sensor network consisting of 15 piezoelectric transducers is designed to cover the entire cylindrical area. The effectiveness is then evaluated experimentally by placing artificial defects on the surface of the COPV. The multi-layered dataset of GW signals was analyzed using both commonly used ultrasonic features (e.g., amplitude, frequency, time of flight) as well as statistical features (kurtosis, skewness, variance, etc.). These features were utilized to compute a damage index, and the effectiveness of the detection performance was assessed using receiver operating characteristic curves. It can be seen that some features are more sensitive and robust under varying experimental conditions. The results show that ultrasonic GW SHM system is a promising solution for damage detection and localization in COPVs.

[DOI: 10.1115/1.4067667]

Keywords: composite overwrapped pressure vessels, guided waves, structural health monitoring, sensors placement, damage detection, damage localization, elastic wave, testing methodologies, ultrasonics

1 Introduction

Composite overwrapped pressure vessels (COPVs) are employed extensively in a diverse range of industries, including automotive and aerospace. With the emergence of compressed natural gas and hydrogen as novel energy sources, their storage is becoming indispensable. In particular, future applications such as hydrogen-powered vehicles and refueling stations, where pressures could reach up to 700 bar are driving this change [1].

The classification of COPVs is dependent on the manner in which the composite fibers are integrated into the design. This results in

three principal types, designated as type II, type III, and type IV, with type I representing a fully metallic vessel [2]. Type III and type IV COPVs results in a notable reduction in overall weight compared to other vessel types. In particular, COPVs exhibit superior energy storage density, rendering them the optimal choice for high-pressure applications. However, when compared to all-metal vessels, COPVs require unique design, manufacturing, and testing specifications [3].

The multilayer nature of the composite laminate used in the construction of the COPVs is associated with a range of potential failure modes [4]. The understanding of these failure modes is at a much more rudimentary level than for metal vessels. The failure mechanisms for COPVs include mechanical damage, stress rupture (composite damage progression), fluid attack, corrosion, fatigue crack growth, liner buckling, manufacturing defects,

¹Corresponding author.

Manuscript received July 22, 2024; final manuscript received November 20, 2024; published online February 5, 2025. Assoc. Editor: Lingyu Yu.

thermal environments, overstress, and external impacts [5]. These failure mechanisms must be controlled to ensure the safe use of COPVs from manufacture to decommissioning.

Safety considerations dictate the need for periodic inspection of these structures. Currently, periodic inspections typically include a hydraulic test along with internal and external visual inspections [6]. However, these manual inspection methods are both time consuming and costly due to the various steps involved, such as mounting and dismounting, process interruptions, hydrogen release, water filling, hydraulic testing, drying, etc. In addition, these approaches increase the risk of external influences on COPVs [7].

In order to overcome the limitations of current inspection strategies, it is possible to introduce alternative inspection schemes that can be performed in situ and during operation. For instance, continuous monitoring through the utilization of structural health monitoring (SHM) systems could facilitate a better assessment of critical damage scenarios by focusing on the entire structure. The integration of such an SHM system could facilitate additional benefits such as damage localization, using the vast amount of structural data for lifetime prediction and for ongoing design improvements to extend the lifetime of the asset.

A number of methods and techniques have been developed and are available for SHM based on different modalities such as vibration, ultrasonic, acoustic emission, eddy current, and others [8]. Among these, ultrasonic guided waves (GWs) are particularly well suited for inspecting various shell-like structures. They can travel long distances along complex structures, and interact with developing defects. A notable example is composite structures, where the material influences the wave behavior, making it possible to detect barely visible damage caused by low-velocity impacts, such as fiber breakage, matrix cracking, and delamination, with just a few transducers. In addition, this method is readily applicable to SHM using lightweight piezoelectric transducers, allowing full coverage of the structure being inspected [3,9].

Over the past few decades, the majority of GW-based techniques employed in the context of cylindrical structures such as pressure vessels have been used for the inspection and assessment of steel pipes. The majority of research conducted on pipe sections has concentrated on the utilization of transducer rings to regulate the excitation of specific modes of cylindrical GWs, such as the axisymmetric longitudinal $L(0,2)$ modes [10,11] or torsional $T(0,1)$ modes [12,13]. Others have configured an active piezoelectric network to capture wave propagation characteristics and the influence of defects on them. Consequently, models and optimization techniques for sensor networks including the optimal distribution of actuating and sensing points were developed to facilitate the collection of more comprehensive data, thereby enabling the identification and evaluation of flaws within the monitored area under study. A novel approach by Ismail et al. [14] presents a method for transforming any complex and closed section surface into a series of flat 2D plates taking into account the continuity of the structure. This involves defining the required boundary conditions to consider wave propagation in various directions. Consequently, the wave propagation characteristics are determined experimentally prior to optimizing the sensor network positions, thereby ensuring that the effect of curvature and other factors on the wave properties is accounted for.

The aforementioned method, in conjunction with the simplification of the COPV to a cylindrical structure, allows for the design of sensor placements that deviate from the conventional forms of rings or axial lines. Optimized sensor arrangements based on GW propagation properties will result in enhanced coverage and more precise damage detection. Nevertheless, the research on damage detection and localization experiments on pressure vessels remains limited. In their study, Li et al. [15] investigated the dispersion characteristics of various GW modes in pressure vessels. Based on the findings of their simulations and experiments, they optimized the excitation center frequency and waveform parameters of GWs. In consideration of the theoretical background of GW in hollow cylinders, Yang et al. [16] put forth an elliptic positioning algorithm

based on coordinate transformation. Moreover, Li et al. [17] proposed a new probabilistic ellipse imaging method to enhance the precision of defect localization on the surface of pressure vessels. While the majority of these studies concentrated on metallic vessels, others investigated the potential of GW for COPV in the context of SHM [18,19]. Memmolo et al. [3] provided a comprehensive overview of the steps required for the design and verification of a GW-based SHM system for defect detection and localization in a COPV.

The principal objective of this paper is to gain insight into the propagation characteristics of GWs in COPV toward the development of a sensor network to provide an optimal coverage of the area under study. This will further enhance the ability to locate and assess damage.

The outline of the paper is as follows: Sec. 2 highlights the methodology to experimentally investigate and evaluate the propagation characteristics of GW in the COPV, with a view to extracting crucial parameters for sensor network design. Section 3 briefly presents the results of the sensor network design optimization, followed by an experimental validation of the network using artificial damages applied on the surface described in Sec. 4. The data are then evaluated with traditional GW tomography fusion techniques using ultrasonic features (time, frequency, amplitude) and statistical features. Finally, the damage visualization capabilities are analyzed.

2 Guided Wave Propagation Characteristics in COPV

2.1 Guided Waves in Cylindrical Structures. In the context of pipe-like structures, GWs are referred to as cylindrical lamb waves [20]. Gazis [21] presented a general solution for the displacement field components of GW particles in long hollow cylinders, thereby establishing the analytical foundation of GWs in a three-dimensional hollow cylinder of infinite range. As a consequence of the cylindrical curvature and dispersion characteristics, the propagated waves display complex multimodal behavior. Multimode GWs are capable of propagating along a variety of paths, including circumferential/flexural, longitudinal, and torsional, which refers traveling through the thickness dimension [22]. Figure 1(a) from Ref. [23] illustrates the various wave mode displacement fields within a bar/pipe.

In cylindrical structures, the GW modes are designated as $L(0,n)$, $T(0,n)$, and $F(m,n)$, as outlined in Ref. [24], which correspond to the longitudinal, torsional, and flexural modes, respectively. The index m denotes the harmonic number associated with the variation in circumference, while the index n represents the order of the wave mode. The modes that span from zero frequency were the first mode. The remaining modes were designated according to the order of their cutoff frequency.

Axisymmetric modes, which include both longitudinal and torsional modes, are of particular interest for damage detection purposes due to their high sensitivity to surface and through-thickness cracks. In particular, the $L(0,2)$ mode is frequently used due to its faster velocity and non-dispersive nature across a broad frequency range [25]. $L(0,2)$ is particularly effective in detecting depth and circumferential changes. Furthermore, Li et al. [15] demonstrated the application of the $L(0,2)$ wave mode in the detection and assessment of leakage in composite pressure vessels with minimal energy loss into the water inside the vessel. Furthermore, $L(0,2)$, which is equivalent to the S_0 mode [26], is less susceptible to attenuations caused by sharp corners [23]. Although $L(0,1)$ typically coexists with $L(0,2)$, it can be excluded from the analysis because it propagates at a slower group velocity. To illustrate, Fig. 1(b) depicts the group velocity dispersion curves for a hollow cylinder made of 30CrMo steel with a diameter of 356 mm and a wall thickness of 5 mm, as calculated using a MATLAB code developed by Kiefer [27].

2.2 Experimental Characterization of the Dispersion Behavior. Given the complex multimodal dispersion behavior, it

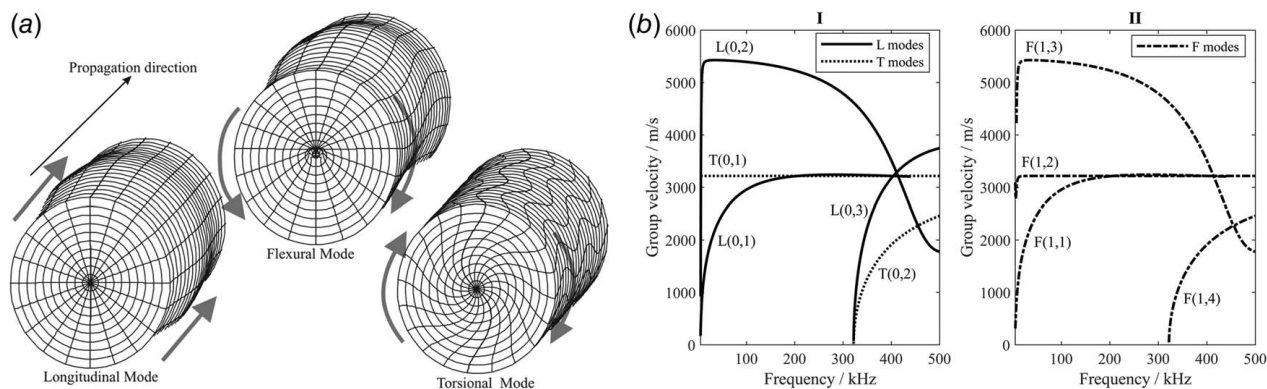


Fig. 1 (a) Displacement fields for different wave propagation modes in a cylindrical structure [23] and (b) group velocity dispersion curves in a hollow steel cylinder with (I) longitudinal and torsional modes and (II) flexural modes

is essential that the design of a damage monitoring sensor network incorporates a precise analysis of the wave propagation in the pressure vessel. This understanding can facilitate the selection of crucial wave parameters, such as the wave mode or the excitation frequency, which are essential for the applicability of damage monitoring [28].

The structure under investigation was a type IV COPV for the storage of hydrogen, manufactured by NPROXX (see Fig. 2(a)). The vessel has a plastic liner with 2 mm wall thickness and over-wrapped carbon fiber-reinforced polymer composite with approximately 16 mm wall thickness. The length of the cylinder between valves was 1510 mm, the cylindrical part was 1270 mm, and the outer diameter was 228 mm. In order to determine the dispersion behavior, an experimental approach was selected. Consequently, a series of seven piezoelectric ceramics (PZTs), positioned at 150 mm intervals, were mounted on the upper surface of the pressure vessel using a thin layer of two-component epoxy adhesive. The PZT numbering is from left to right, with the numbers increasing from 1 to 7, as shown in Fig. 2(a). Two PZTs were positioned at 90 deg and 180 deg to PZT 4 (numbers 8 and 9, respectively) along the midpoint of the pressure vessel. Furthermore, at the line of PZT 7, two additional PZTs were positioned at 90 deg and 180 deg (numbers 10 and 11, respectively). DuraAct patch transducers (P-876K025) with a 10 mm circular ceramic embedded in a ductile polymer from PI Ceramics (Lederhose, Germany) were employed as sensors. To examine the wave propagation a tone burst with five cycles was generated using a multifunction data acquisition system (NI-USB 6366), then amplified to 300 V_{pp} using a high-voltage amplifier (Falco Systems WMA-300). This was used as the excitation signal. While one PZT served as the actuator, the remaining PZTs functioned as receivers, with the role alternating until all the PZTs were used as actuators. A series of

excitation frequencies was employed ranging from 10 kHz to 250 kHz in increments of 5 kHz or 10 kHz. The dispersion curves were generated using the measurement results obtained from the sender-receiver combination PZT 3 to PZT 5 (spaced at 300 mm). For each of the excitation frequency, the wave packages (modes) were identified and their group velocities were calculated based on the first, second, or third peak of the sensed signals depending on their clarity. This was done using the time of flight and the known distance between the exciter and the receiver (Fig. 3).

Four modes were identified: modes 1 and 2 are traveling at a slower speed than modes 3 and 4. Modes 1 and 2 were observed to have a nearly constant speed as the frequency increased, similarly for modes 3 and 4 above 180 kHz. In the frequency range below 180 kHz, modes 3 and 4 exhibited a notable variation in the group velocity, as illustrated in Fig. 3(a). This reached a peak value of approximately 6000 m/s at 140 kHz. An investigation into the amplitude of the four modes identified revealed that modes 3 and 4 exhibited a higher amplitude than modes 1 and 2, as illustrated in Fig. 3(b). The amplitudes of modes 1 and 2 were observed to be markedly low above 50 kHz. Modes 3 and 4 exhibit two peaks, the first around 130 kHz (with multiple modes present) and the second at 220 kHz (with one dominant mode).

In addition to the measurements obtained with the PZT sensor array, line scans were conducted using a scanning laser Doppler vibrometer (SLDV), as shown in Fig. 2(b). This method allows for the acquisition of a high spatial resolution of the propagating wave field, thereby providing a higher density of information regarding the detection and analysis of the wave modes. The wave field was measured along a scan direction using a chirp signal with a frequency range of 20–500 kHz and a duration of 200 μs as excitation, as previously described by Kudela et al. [29]. The generated chirp signal was fed to an arbitrary wave

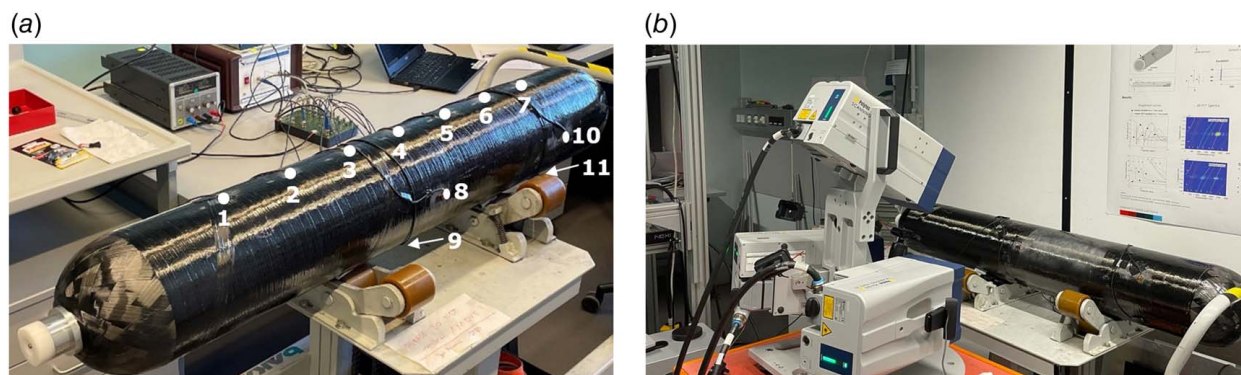


Fig. 2 (a) COPV with sensor network for examining the propagation of GWs and (b) experimental setup for investigating the propagation of GWs in a COPV using an SLDV

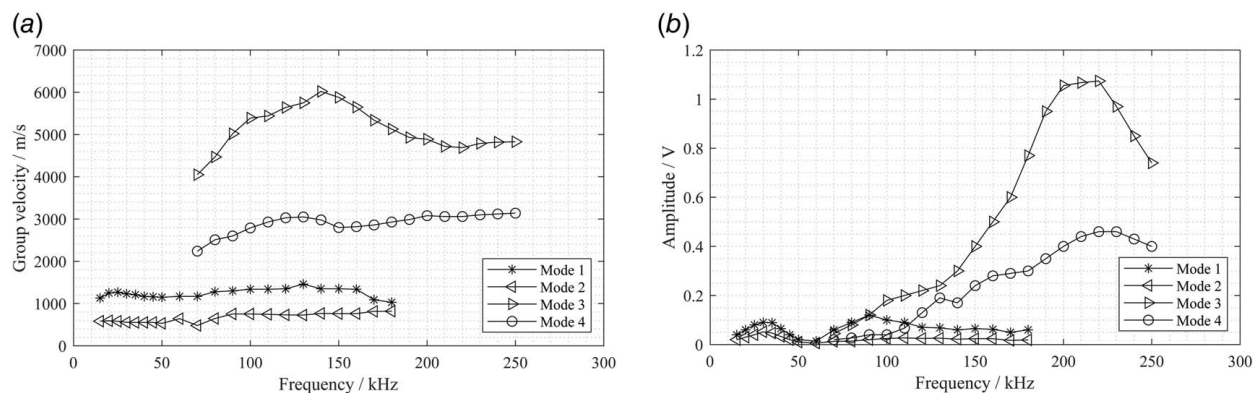


Fig. 3 Dispersion curves using experimental data based on data collected using PZT 5 (sensor) when exciting with PZT 3: (a) group velocity and (b) amplitude

generator (Agilent 33521 A) and amplified to 100 V using a high-voltage amplifier (Ciprian HVA-400-A) prior to being fed to PZT 8. The wave field was recorded using a 3D SLDV (PSV-500-3D-HV, Polytec).

Subsequently, the acquired temporal and spatial data can then be transformed into the frequency–wavenumber domain (f – k plot) using two-dimensional fast Fourier transform (2D FFT), which allows the individual modes to be identified. Figure 4(a) illustrates the outcome of the SLDV scan conducted along the longitudinal axis. A visual examination of the diagram reveals the presence of two distinct wave modes, characterized by a low propagation speed and frequency. In contrast, two additional modes are more prominent and propagate at a higher speed. The four modes in question can be matched with modes 1–4, which were previously identified in the preceding experiment. The dispersion curves for the phase velocity can be created using the relationship $v_p = 2\pi f/k$, as illustrated in Fig. 4(b) by means of the f – k plot.

2.3 Laser Line Scan Results for Different Directions. There is a strong dependence between the mechanical properties of the materials and the lay-up sequence of the fibers. As the mechanical properties of the material are dependent on the direction or orientation of the fibers, the characteristics of wave propagation are also subject to change. In order to gain a deeper insight into the excited modes in the COPV, a laser line scan was conducted along the longitudinal, circumferential, and diagonal directions using the aforementioned setup with the SLDV. Furthermore, the response of the PZT wafers positioned around the circumference is investigated to gain deeper insight into the relationship between direction and wave propagation characteristics in the COPV.

Figure 5 illustrates the f – k images captured in the three scanning directions. Additionally, it illustrates the combinations of the various laser measurement component (in X , Y or Z) to generate

the f – k images. The combination of the laser measurement components was explored to understand their relation to the propagation direction of the modes. In particular, the question was whether the modes were in-plane or out-of-plane. This approach facilitates the enhanced visualization of specific modes in accordance with their intrinsic characteristics. No weighting factors were employed when combining the measurements obtained from the various components. Each laser head captures different modes (in-plane or out-of-plane) and as such the information from each measurement is inherently different and does not require additional weighting. The SLDV was orientated in a fixed position relative to the test object. The measurements were constrained by the scanning capabilities of the instrument, which were limited to a range of less than 100 mm. This limitation was primarily due to the curvature of the object, and the objective was to maintain a relatively narrow scanning range by measuring only along the available circumferential and diagonal lines. As can be observed in Fig. 5, the presence of two dominant modes is evident. In comparison with the velocities illustrated in Fig. 4, it is probable that these represent mode 3 and 4. Furthermore, the slope of the f – k plot is consistent regardless of the direction of the SLDV scan (illustrated by the green and red arrows), indicating that the phase velocity remains unaltered when the scan direction is varied. This may indicate the presence of quasi-isotropy in the materials. Further analyses were conducted to investigate the dependency of the propagation direction. The responses from PZT 4, 8, and 9 were analyzed while exciting using PZT 2. PZT 2 and PZT 4 are located on the surface of the vessel and spaced at 300 mm, PZT 8 is placed on the side, and PZT 9 is on the bottom side directly below PZT 4 (see Fig. 2(a)). Figure 6 illustrates a typical response at an excitation frequency of 220 kHz.

The propagating speed of the first transmitted wave was determined to exhibit a range of approximately 5000 m/s, contingent on the specific PZT sensor employed. A maximum variation of

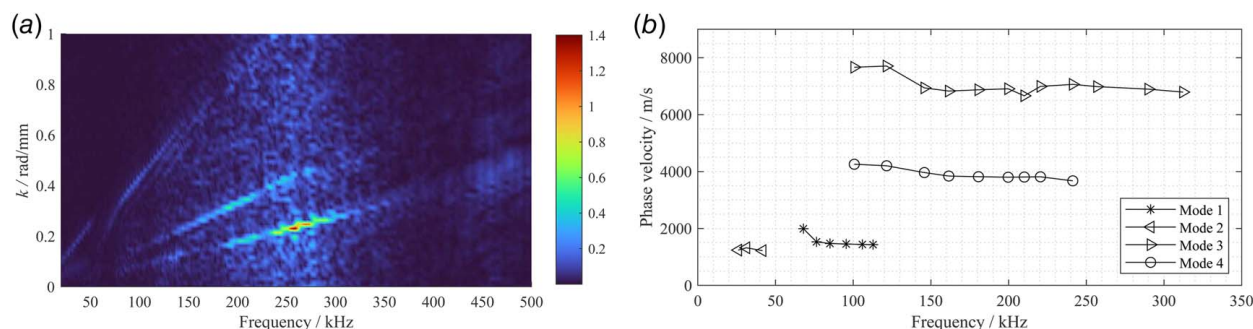


Fig. 4 f – k diagram of the SLDV line scan along the longitudinal direction: (a) excitation using a chirp signal with a frequency range of 20–500 kHz and (b) phase velocities extracted from (a) plotted as dispersion curves

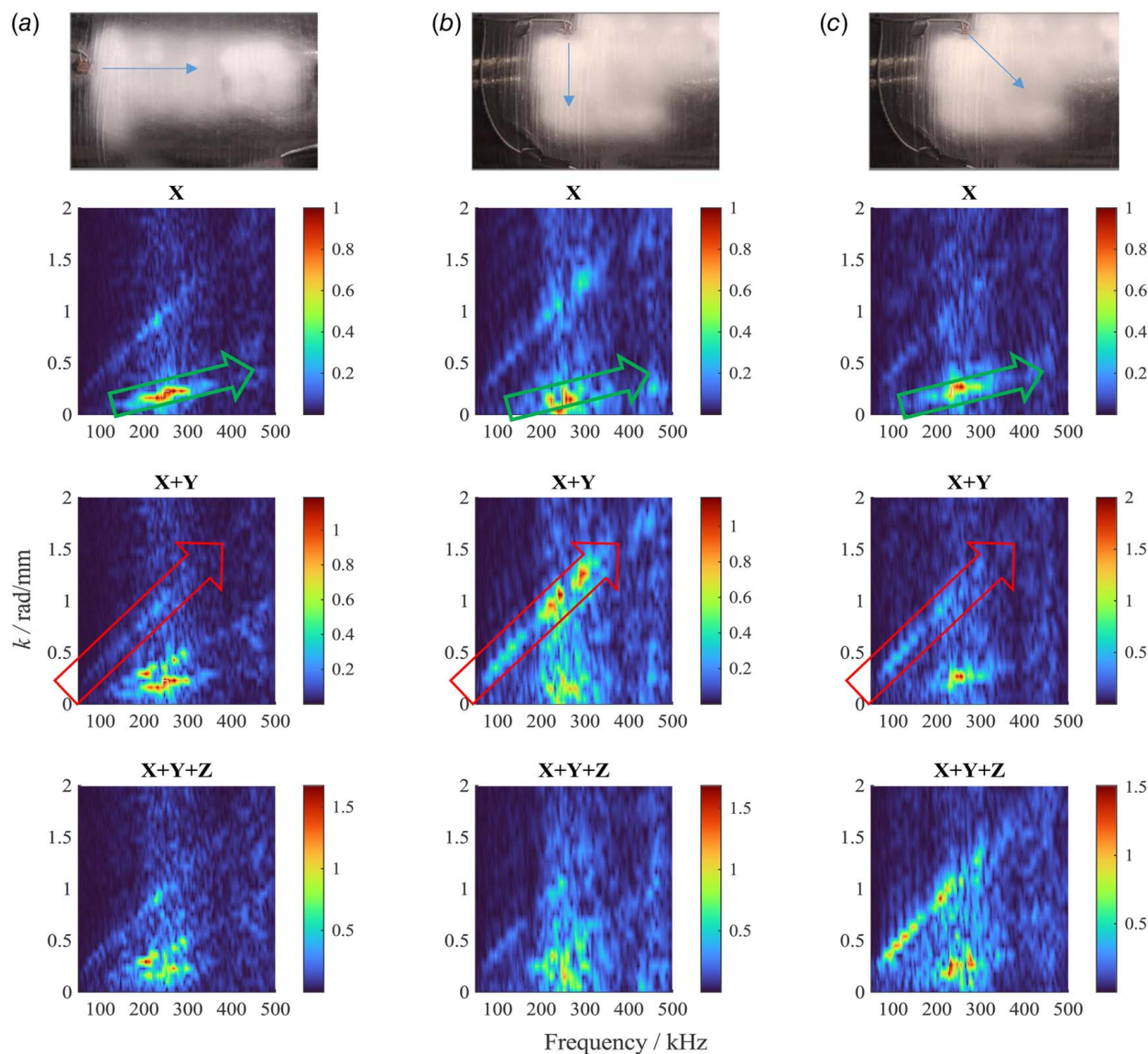


Fig. 5 f - k diagrams for the chirp excitation in various directions along the (a) longitudinal, (b) circumferential, and (c) diagonal directions

2% was observed in the propagating speed. Similarly, when other excitation frequencies were employed (see Table 1), the velocity along the various directions exhibited minimal variation, with the exception of the 180 kHz excitation frequency. At 180 kHz, the variation reached 27%. It is likely that this outlier is the result of a measurement error.

2.4 Guided Wave Propagation at Selected Frequencies.

Following an investigation into the wave propagation behavior of GW in the COPV, it is essential to ascertain the quality of the signal and its interaction with any potential defects. It is of significant importance to select an excitation frequency that exhibits a minimal number of wave modes in order to facilitate straightforward signal processing. For this reason, the GWs were excited and captured at various locations on the surface of the COPV, in accordance with the methodology outlined in Sec. 2.2. A range of excitation frequencies was considered in this experiment, spanning from 10 kHz to 250 kHz. Figure 7(a) illustrates the captured responses along the linear array when PZT 1 was actuated at 130 kHz. The black dotted line presents the Hilbert transform of the data, facilitating the extraction of the time of flight. Mode 3 was

the predominant transmission mode as the wave propagated along the length of the vessel. Furthermore, the velocity of the first transmission remained consistent, with the group velocity increasing from 5000 m/s to 5800 m/s. The results for an excitation frequency of 220 kHz are presented in Fig. 7(b). Mode 3 was also the dominant mode, although its velocity was slightly lower, reaching 4690 m/s. The 130 kHz and 220 kHz were investigated in greater depth due to their prominence in the wave dispersion curves previously generated. Therefore, the following section will investigate the sensitivity of these models to defects or simulated damage, before proceeding to the design of the sensor network.

2.5 Mode Selection—Attenuation and Sensitivity to Simulated Damage.

The experimental schemes are illustrated in Fig. 8. In Fig. 8(a), the first PZT wafer (PZT 1) served as the actuator in this scenario while the remaining wafers were designated as receivers. The last PZT is situated at a distance of 900 mm away from the actuator. To ascertain the impact of excitation frequency on the attenuation curves of the material, a series of tests were conducted, evaluating the amplitude of the first peak received at each sensor as a function of distance from the actuator.

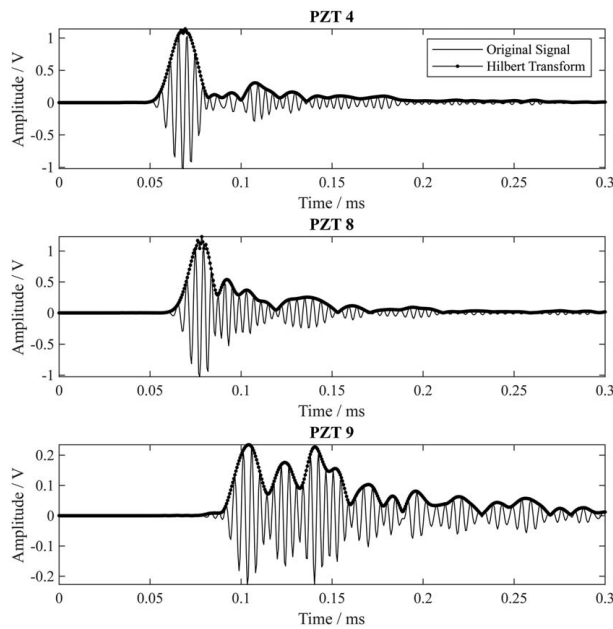


Fig. 6 Effect of the propagation direction on the mode of interest group velocity, here at 220 kHz. Sending with PZT 2 and receiving with PZT 4, 8, and 9, respectively.

Table 1 Variation of group wave velocity with direction

Frequency (kHz)	PZT 4 (m/s)	PZT 8 (m/s)	PZT 9 (m/s)	% variation PZT 4/ PZT 8	% variation PZT 4/ PZT 9
140	5691	6140	5402	7.89	5.07
160	5930	5837	5412	1.58	8.74
180	4440	5645	4845	27.12	9.11
200	6143	6104	5924	0.63	3.56
220	5152	5199	5060	0.91	1.80
240	5259	5269	5141	0.20	2.24

It was observed that for the excitation frequencies of 110 kHz and 130 kHz, the wave can propagate for 900 mm while still conserving approximately 30% of its amplitude (Fig. 8(a)—bottom). Excitation

at a higher frequencies (150 kHz and 220 kHz) resulted in a significant loss of amplitude, with 90% reduction observed after 900 mm propagation distance. However, the signal-to-noise ratio (SNR) remained within an acceptable range. It can therefore be assumed that a propagation distance of 900 mm will ensure that the initial transmission is still pronounced.

In order to ascertain the extent of coverage around the line of sight, two PZT elements were positioned at a distance of 450 mm from each other. The wave was actuated with one PZT, utilizing an excitation frequency of 220 kHz, with the other PZT serving as the receiver. A steel block with a mass of 300 g and a diameter of 30 mm was used to simulate damage. It is anticipated that this will result in a reduction in the amplitude of the wave as it propagates along the path. The block was initially positioned along the path centerline and subsequently displaced from this position in a gradual manner, as illustrated in Fig. 8(b)—top. As the damage was displaced from the path centerline, the sensor was employed to ascertain the amplitude of the propagating wave. The bottom figure of Fig. 8(b) displays the first transmission signals for a variety of damage scenarios. The various scenarios correspond to a gradual shift of the block center away from the path centerline, with distances of 0 mm (damage case 1), 30 mm (damage case 2), and 45 mm (damage case 3) being considered. When the damage was situated at 45 mm (damage case 3), the amplitude reduction was 40%. The reduction in the amplitude of the wave signal beyond this point is minimal. Therefore, the path coverage was set to 45 mm on either side of the path.

3 Design of Sensor Network for Damage Detection

In a previous study by Ismail et al. [14], a model was developed to optimize the placement of PZT wafers on any shape with closed sections, including curves. The methodology employed for the optimization of PZT networks on closed sections entailed the unfolding of the original surface to yield a rectangular shape of dimensions $P \times L$. In this approach, the length of the closed section L and the circumference length P were used to determine the unfolding of the original surface (see Fig. 9(a)).

The objective of the proposed optimization algorithm was to maximize the coverage of a predefined set of control points on the unfolded cylinder, while simultaneously minimizing the number of PZT wafers. Accordingly, the structure was discretized to an adequate number of control points representing the structure geometry. The coverage level (n), which is a parameter selected according to user-specific requirements, was defined as the

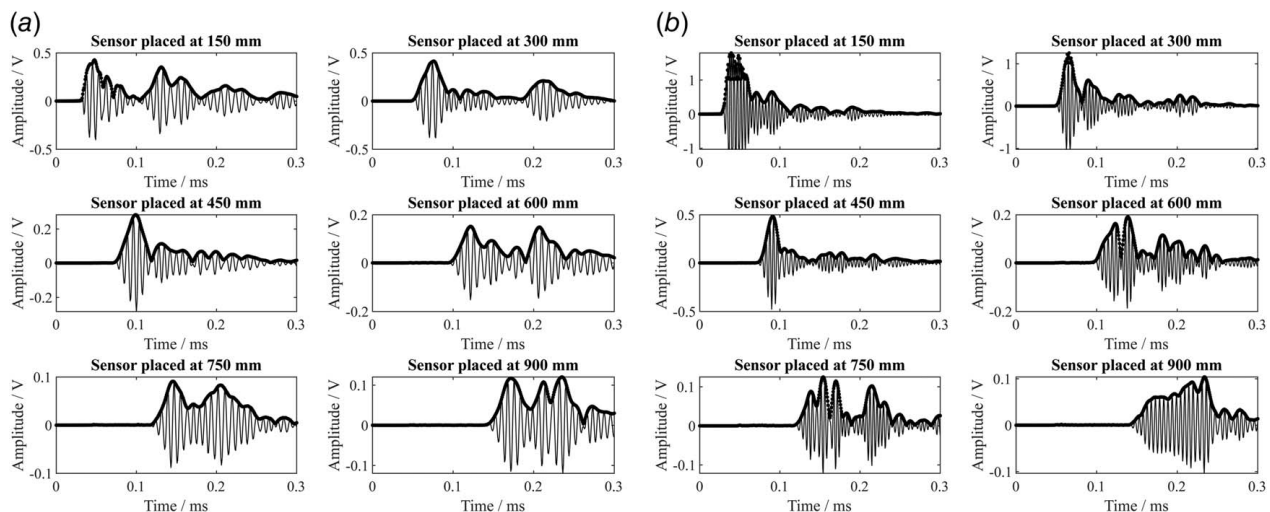


Fig. 7 Sensor responses captured at various sensors positions away from actuator 1 using an excitation frequency of (a) 130 kHz and (b) 220 kHz. The black dotted line indicates the Hilbert transform of the corresponding signals.

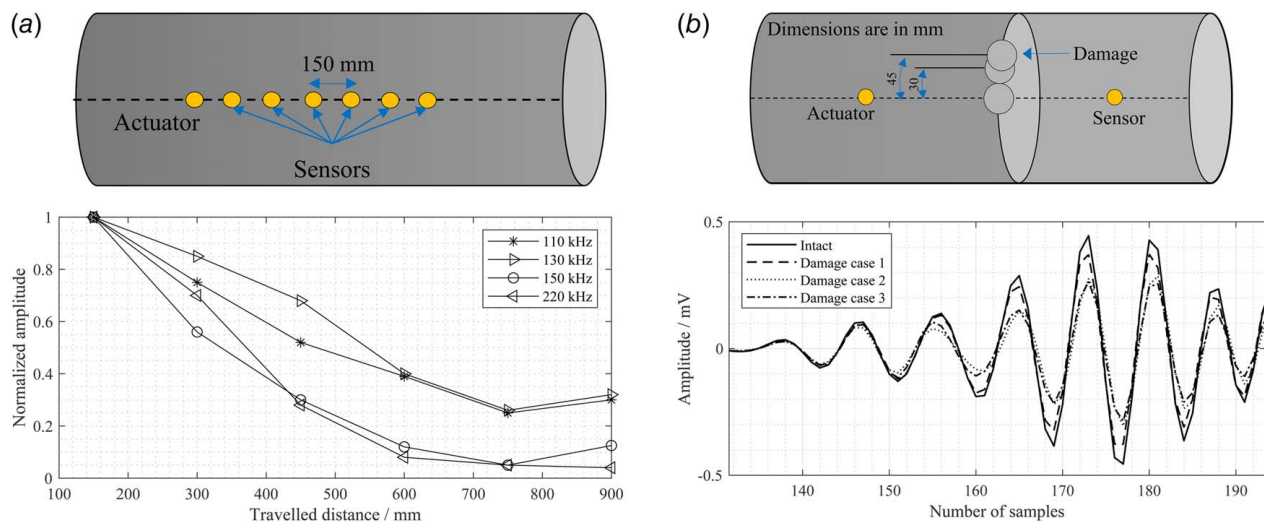


Fig. 8 Experimental scheme for the wave propagation distance: (a)—top—with attenuation curves over traveled distance using various excitation frequencies—bottom—and the path coverage experiment; (b)—top—with the first transmission signals of the 220 kHz excitation and their sensitivity to different simulated damage scenarios—bottom

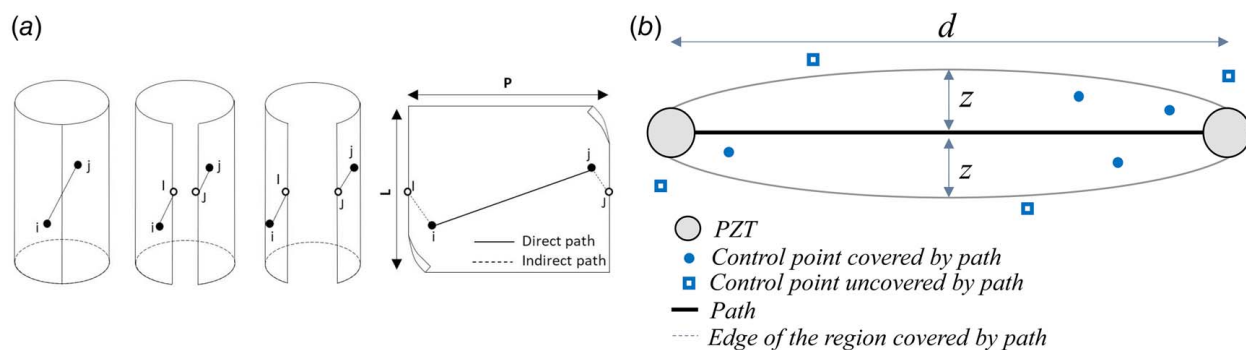


Fig. 9 (a) Unfolding a closed section with resulting sensing paths [14] and (b) sketch to illustrate the constraints and terminology of the optimization model

number of sensing paths necessary to cover a control point. Huang and Tseng [30] demonstrated that accurate damage localization necessitates coverage with three sensing paths ($n = 3$). A sensor path is defined as a sender-receiver combination with a maximum distance d and a path coverage z . The term path coverage z is used to describe the distance from the centerline of a path at which a control point is still considered to be covered. In this case, z is equal to 45 mm. The maximum distance, d , accounts for the attenuation of the propagating signal over the distance at which a sufficient signal-to-noise ratio is still present. This value was determined through prior experimentation and was found to be equal to 900 mm. Moreover, a minimum angle between two sensor paths ($\alpha_{\min} = 10$ deg), and a minimum distance between two sensors ($d_{\min} = z = 45$ mm) are defined as boundary conditions. An illustration of the terminology and boundary conditions is provided in Fig. 9(b).

3.1 Sensor Network Design Optimization. In the design of the sensor network, two principal approaches may be considered:

- (1) The generation of a preliminary solution (either uniform or random) should be followed by the optimization of the coverage in order to achieve the desired level of coverage, as determined by the developed model. This may result in the random placement of the PZTs.
- (2) Definition of a solution and evaluation of coverage based on the developed model.

In this study, the second approach was employed, as a uniform solution was deemed to be of interest. The uniform solution is characterized by a more organized placement of the sensors, thereby facilitating the integration of additional sensing modalities on the pressure vessel, such as acoustics and fiber optic sensing.

On this basis, three designs were evaluated, each consisting of a varying number of PZT wafers: 9, 12, and 15, respectively. Each of the designs comprised a three-ring structure, with a distance of 600 mm between each ring along the length of the vessel. The number of PZTs was distributed in a uniform manner between the rings and positioned at equal intervals along the circumference. Figure 10(a) illustrates the initial design, comprising nine PZT elements. The figure on the left depicts the sensing paths, while the one on the right illustrates the control points that have been covered (depicted as a blue dot) and those that remain uncovered (depicted as a blue square). This design resulted in an overall coverage of 19% of the total control points. An increase in the number of PZTs from 9 to 12 resulted in a notable enhancement in the coverage, reaching 81.5%. This is illustrated in Fig. 10(b). An additional increase in the number of PZTs resulted in 100% coverage, as illustrated in Fig. 10(c). In accordance with the aforementioned, design No. 3 will be adopted and subjected to further experimental validation in the following section.

4 Damage Detection and Localization Performance

4.1 Experimental Validation. In order to test the sensor network, the optimized design, comprising three rings with five

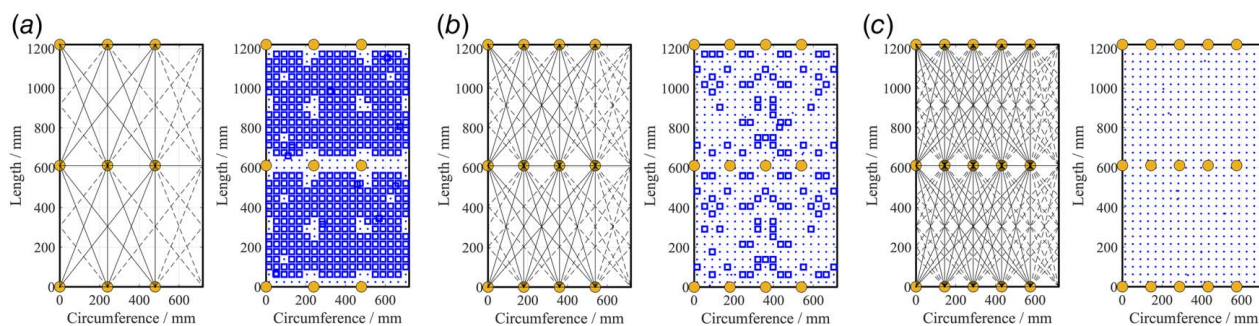


Fig. 10 Design for sensor network with (a) 9, (b) 12, and (c) 15 PZT wafers evenly distributed around the vessel in three separate rings

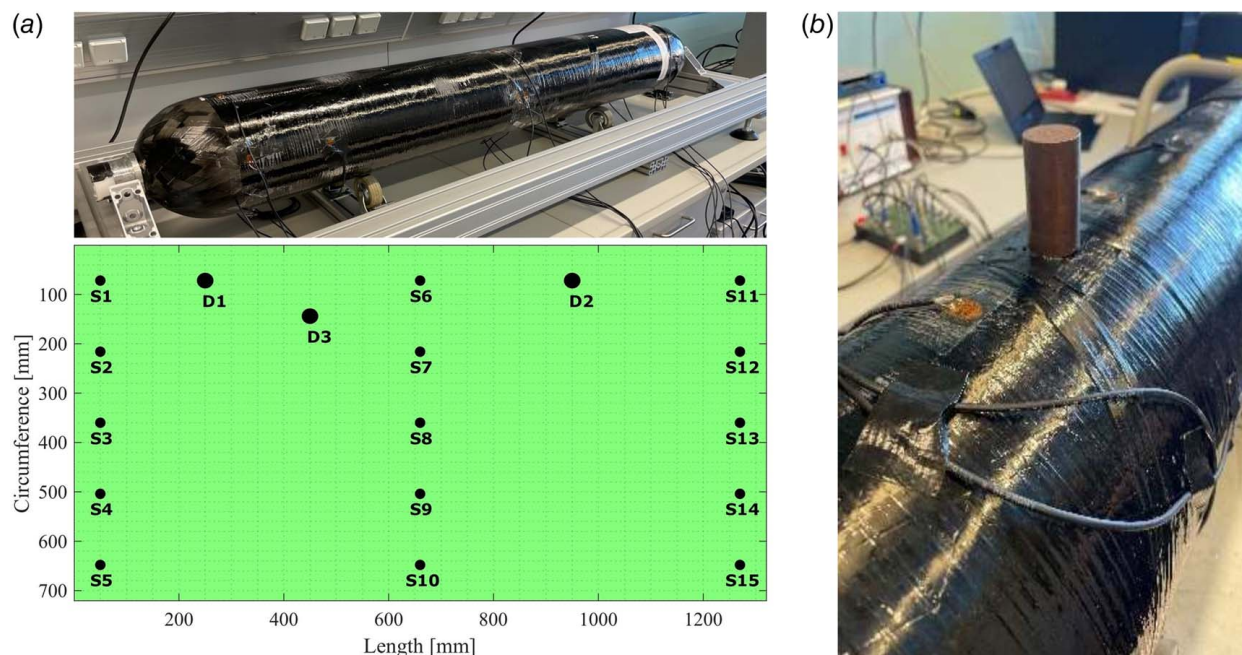


Fig. 11 (a) Unwrapped cylindrical surface of COPV with the sensor network's PZT positions 1–15 and the artificial damage locations D1–D3 and (b) photograph of a metal block used as artificial damage

sensors each, was initially transferred to the COPV. As in the preceding experiments, PZTs (P-876K025) were utilized, which were applied to the surface with a thin layer of two-component epoxy adhesive. The distance between the PZTs in the circumferential direction was 144 mm, while the distance between the individual rings was 600 mm. A tone burst comprising five cycles and an amplitude of 100 V_{pp} was employed to excite the GWs within the pressure vessel. The data acquisition (Verasonics Vantage 64 LF) was conducted using the pitch-catch method, with a single PZT serving as the actuator and the remaining ones functioning as receivers. This process was repeated until all the PZTs had served as actuators in turn (round-robin).

In order to evaluate the robustness of the sensor network in terms of damage detection, a series of tests were conducted in which a variety of artificial defects and defect combinations were applied to the COPV. In essence, five different artificial damage scenarios (D1–D5) were investigated. Metal blocks as artificial damages were placed in various parts of the network to cover a wider range of scenarios and ensure that the network provided full coverage of the area under study (Fig. 11). D4 was a combination of D2 and D3, and D5 was a combination of D1 and D3. The pressure vessel was supported in two ways, which were categorized as the boundary condition. First, the apparatus was affixed to casters in

the initial and final thirds of the COPV, with four supporting points. Subsequently, the COPV nozzles were positioned on a frame to assume a laid-free condition. Two distinct excitation frequencies were selected, namely $f_1 = 130$ kHz and $f_2 = 220$ kHz,

Table 2 Experimental setup variations to evaluate the robustness of the sensor network in terms of damage detection and localization

Types of variations	Description
Damage scenarios	D1: Between S1 and S6, 20 cm away from S1 D2: Between S6 and S10, 30 cm away from S6 D3: Between S1, S6, and S7, 40 cm and 36 deg from S1 D4: Combination of D2 and D3 D5: Combination of D1 and D3
Damage sizes	DS1: Metal block weighting 513 g, 3 cm diameter DS2: Metal block weighting 330 g, 3 cm diameter
Boundary conditions	BC1: COPV on fixed castors, support on four points BC2: COPV assumed laid free, nozzles on the frame
Excitation frequencies	F1: 130 kHz F2: 220 kHz

Table 3 DI based on feature extraction

Damage index	DI	Formula
Correlation coefficient	CC	$\rho = \frac{\sum_{i=1}^N S_{H,i} S_{D,i} - \sum_{i=1}^N S_{H,i} \sum_{i=1}^N S_{D,i}}{\sqrt{\sum_{i=1}^N S_{H,i}^2 - (\sum_{i=1}^N S_{H,i})^2} \cdot \sqrt{\sum_{i=1}^N S_{D,i}^2 - (\sum_{i=1}^N S_{D,i})^2}}$
Frequency domain maximum amplitude	FAMAX	$DI = \frac{ f_{\max}^H - f_{\max}^D }{f_{\max}^H}, f_{\max}^{H,D} = \max[S_{H,D}(f)]$
Skewness of frequency domain response	F-SKEW	$DI = \frac{ f_{\text{skew}}^H - f_{\text{skew}}^D }{f_{\text{skew}}^H}, f_{\text{skew}}^{H,D} = \text{skewness}[S_{H,D}(f)]$
Skewness	SKEW	$DI = \frac{ t_{\text{skew}}^H - t_{\text{skew}}^D }{t_{\text{skew}}^H}, t_{\text{skew}}^{H,D} = \text{skewness}[S_{H,D}(t)]$
Signal sum of squared differences	SSSD	$DI = \frac{\sum (S_H - S_D)^2}{\sum S_H^2}$
Hilbert transform-based TOF percentage difference	HTOF	$DI = \frac{ t_{\max H}^H - t_{\max H}^D }{t_{\max H}^H}, t_{\max H}^{H,D} = \arg\max(\mathbf{H}[S_{H,D}(t)])$
Area under frequency spectrum	FAUC	$DI = \frac{ f_{\text{auc}}^H - f_{\text{auc}}^D }{f_{\text{auc}}^H}, f_{\text{auc}}^{H,D} = \text{trapz}[S_{H,D}(f)]$
Kurtosis of frequency domain response	F-KURT	$DI = \frac{ f_{\text{kurt}}^H - f_{\text{kurt}}^D }{f_{\text{kurt}}^H}, f_{\text{kurt}}^{H,D} = \text{kurtosis}[S_{H,D}(f)]$
Kurtosis	KURT	$DI = \frac{ t_{\text{kurt}}^H - t_{\text{kurt}}^D }{t_{\text{kurt}}^H}, t_{\text{kurt}}^{H,D} = \text{kurtosis}[S_{H,D}(t)]$
Cross-correlation maximum percentage difference	CCMPD	$DI = \frac{ \max[CC_H] - \max[CC_D] }{\max[CC_H]}, t_{\max CC}^{H,D} = \arg\max([CC_{H,D}(t)])$
Cross-correlation-based TOF percentage difference	CCTOF	$DI = \frac{ t_{\max CC}^H - t_{\max CC}^D }{t_{\max CC}^H}, t_{\max CC}^{H,D} = \arg\max([CC_{H,D}(t)])$
Signal amplitude Hilbert transform maximum	SAHM	$DI = \frac{ \max[\mathbf{H}[S_H]] - \max[\mathbf{H}[S_D]] }{\max[\mathbf{H}[S_H]]}$
Parametric resonance frequency differences	ARRFD	$DI = \frac{ f_{\text{peak}}^H - f_{\text{peak}}^D }{f_{\text{peak}}^H}, f_{\text{peak}}^{H,D} = \arg\max[S_{H,D}(f)]$
Variance of frequency domain response	F-VAR	$DI = \frac{ f_{\text{var}}^H - f_{\text{var}}^D }{f_{\text{var}}^H}, f_{\text{var}}^{H,D} = \text{variance}[S_{H,D}(f)]$
Variance	VAR	$DI = \frac{ t_{\text{var}}^H - t_{\text{var}}^D }{t_{\text{var}}^H}, t_{\text{var}}^{H,D} = \text{variance}[S_{H,D}(t)]$

due to their high energy dominance in the dispersion curves and the clear separation of the propagating modes within the designated frequency range (Figs. 3 and 4). The various combinations of experimental setup variations are outlined in Table 2. Prior to the examination of various artificial damage scenarios, a baseline measurement of the pristine state was conducted to facilitate the calculation of the difference between the damage-introduced state and its subsequent influence.

4.2 Reconstruction for Probabilistic Damage Localization.

The network of 15 PZT elements in the panel provides 210 sensing paths. Due to the dual functionality of the PZT elements

and the absence of nonlinearity, the number of paths can be reduced to 105 distinct paths (for example, instead of utilizing both paths 1-6 and 6-1, only path 1-6 is considered). It is anticipated that the damage will be located at the intersection of the most severely damaged paths, that is to say, the paths exhibiting the highest anomaly measures (the greatest deviation from the pristine state). The damaged image was reconstructed by dividing the monitored zone (cylindrical part) into a uniform grid of square cells, each with an area of 1 mm². The presence of damage in each cell was evaluated by aggregating the anomaly measures from all the sensing paths contributing to that cell.

The recorded signals are processed based on a reconstruction algorithm for probabilistic inspection of damage (RAPID), as

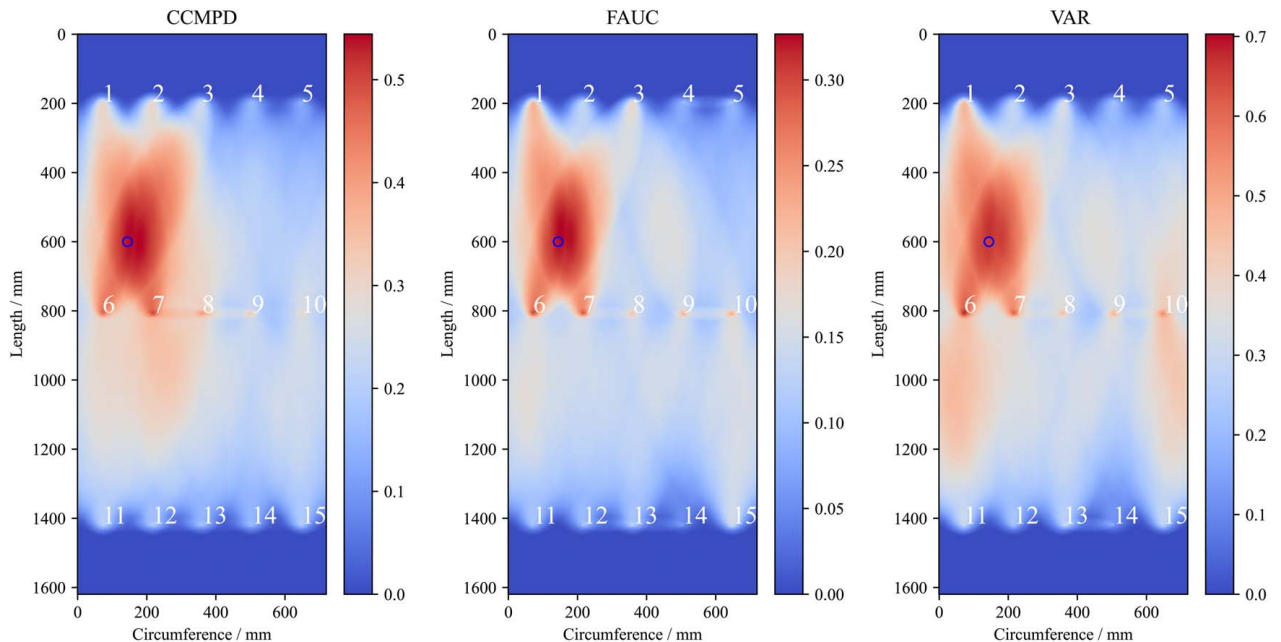


Fig. 12 RAPID results for damage scenario D3_DS2_BC2_F2 using damage indicators CCMPD, FAUC, and VAR

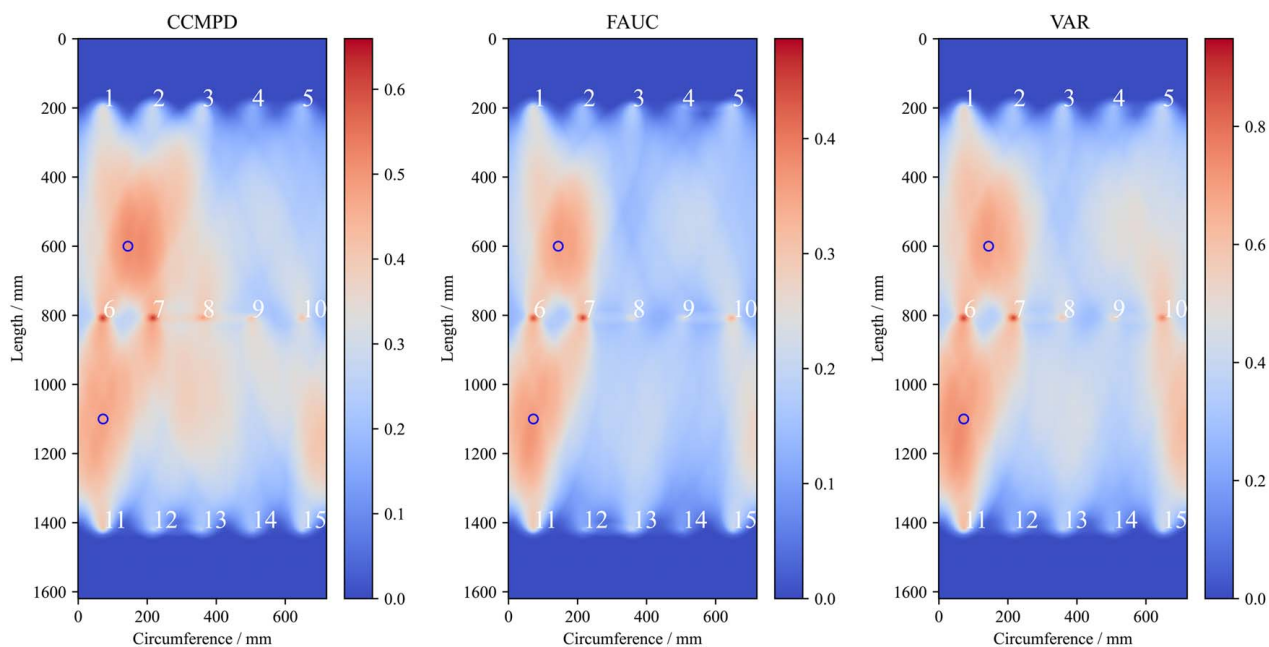


Fig. 13 RAPID results for damage scenario D4_DS3_BC1_F2 using damage indicators CCMPD, FAUC, and VAR

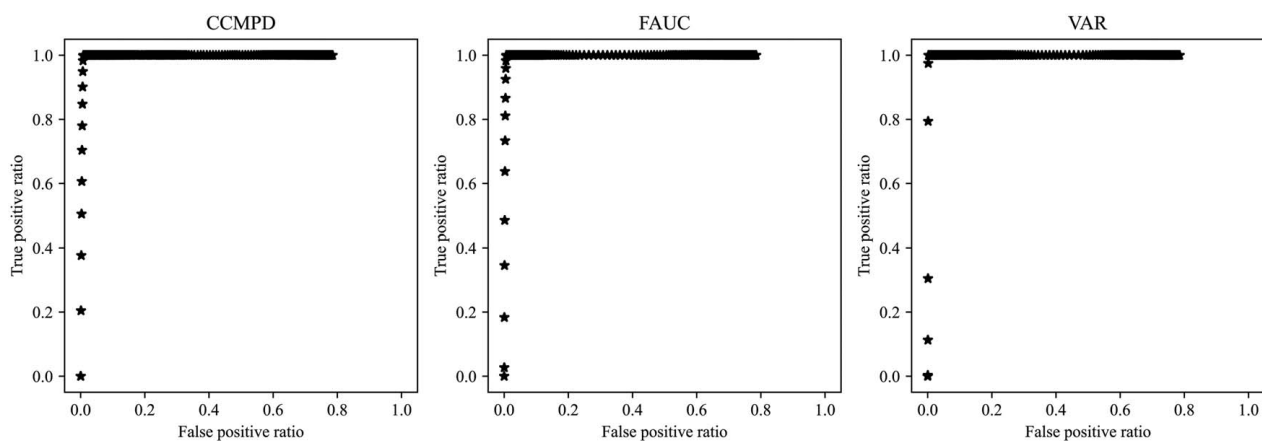


Fig. 14 ROC curve for damage scenario D3_DS2_BC2_F2 using damage indicators CCMPD, FAUC, and VAR

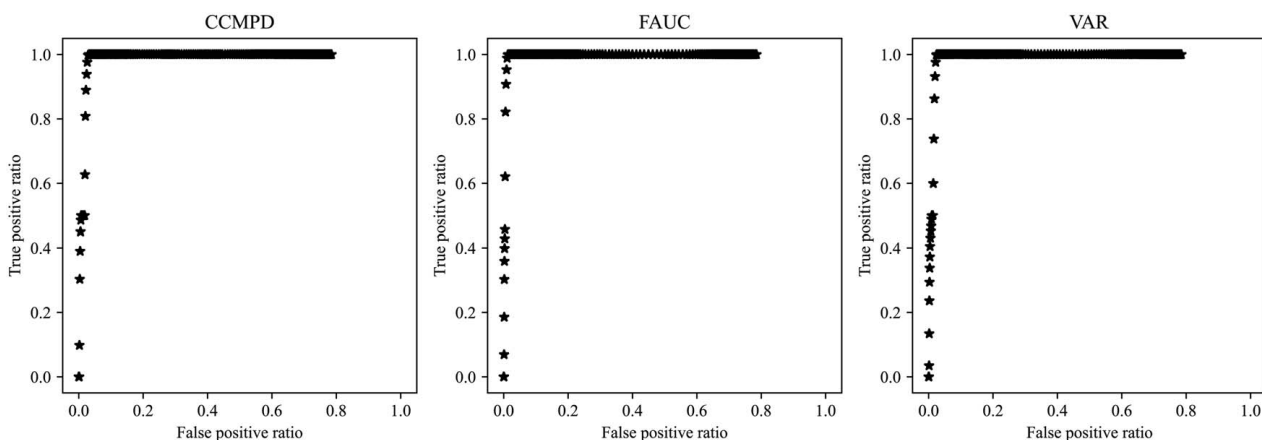


Fig. 15 ROC curve for damage scenario D4_DS3_BC1_F2 using damage indicators CCMPD, FAUC, and VAR

outlined in Ref. [31]. In order to represent the shortest possible sensor path, virtual sensors were considered in the reconstruction algorithm. The damage intensity, denoted by $I(x)$, is calculated at each point (x, y) by taking into account the influence of actuator-sensor paths. The intensity indicates the probability of a damage occurrence (i.e., the higher the value, the more likely damage is present), and is calculated as follows:

$$I(x, y) = \sum_{k=1}^{N_p} (1 - \rho_k) \left(\frac{\beta - R(x, y)}{\beta - 1} \right) \quad (1)$$

where ρ_k is the correlation coefficient of the k th actuator-sensor path, N_p is the number of paths, β is the scaling factor determining the area of influence (the higher the bigger area of influence on wider access), and $R(x, y)$ defined as

$$R(x, y) = \begin{cases} \frac{\sqrt{(x-x_i+\alpha(x_i-x_j))^2 + (y-y_i+\alpha(y_i-y_j))^2} + \sqrt{(x-x_j+\alpha(x_j-x_i))^2 + (y-y_j+\alpha(y_j-y_i))^2}}{\sqrt{(x_i-x_j-2\alpha(x_i-x_j))^2 + (y_i-y_j-2\alpha(y_i-y_j))^2}} \\ \beta \text{ for } R(x, y) \beta \end{cases} \quad (2)$$

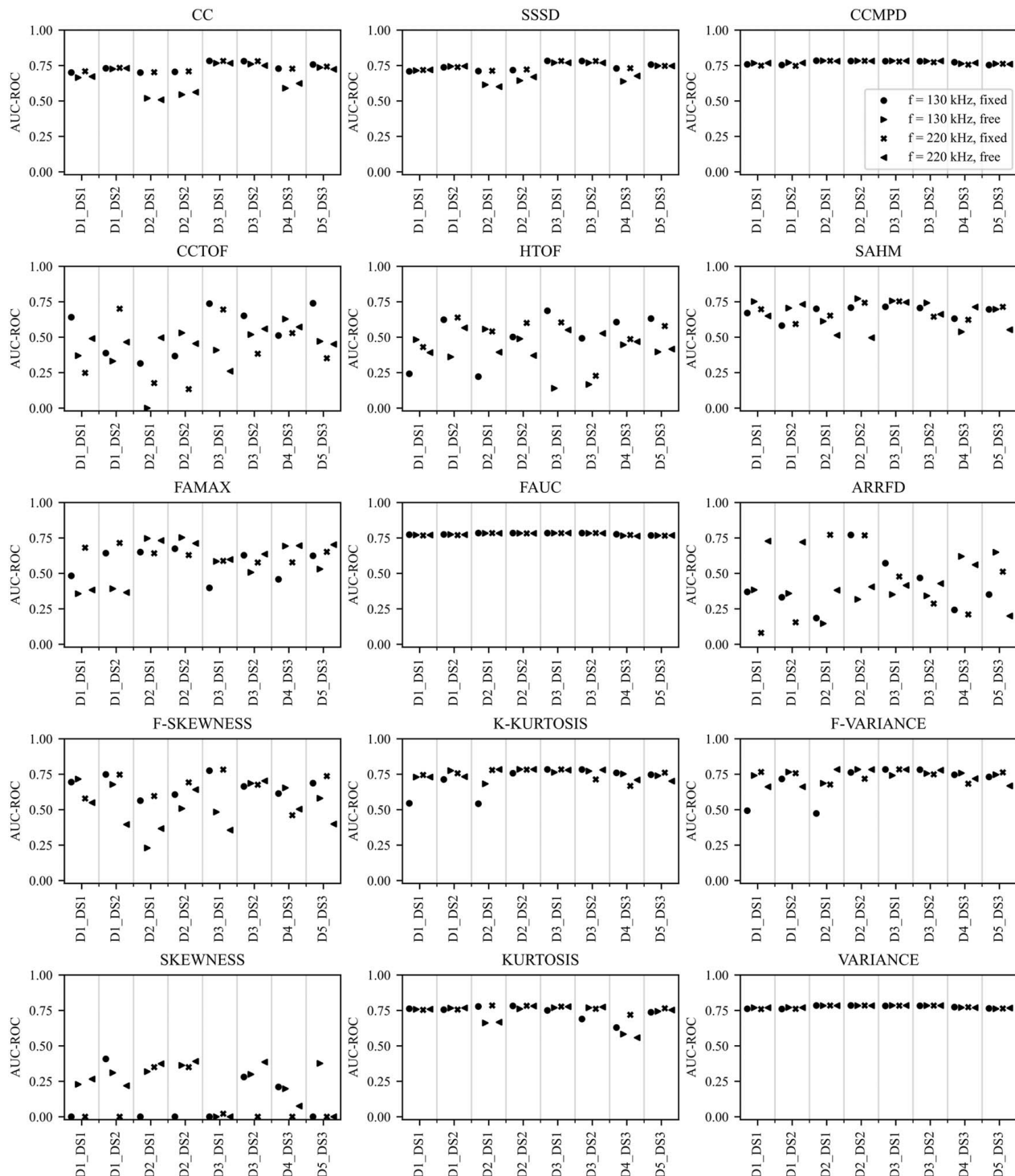


Fig. 16 Detection performance of DIs based on AUC-ROC curve for eight different experimental conditions

where α is the scaling factor determining the area of influence on the actuator-sensor path. Based on the effective area of GW propagation through the pressure vessel, the values of α and β were selected as 1 and 1.05, respectively. In order to enhance the effective zone for the distribution function, it is necessary to augment both the values of α and β .

There are more alternative methods for calculating a damage index (DI) that can be used to indicate the presence of a damage, rather than relying on the correlation coefficient [32]. In this study, the performance of 15 distinct DIs is assessed based on their response characteristics over time and frequency, as well as their shape characteristics. These are presented in Table 3 along with their corresponding mathematical formulas. The DIs are typically defined as the calculated difference between the pristine state (H) and the damage state (D). The following relationship exists between the value of the correlation coefficient and the DI:

$$\rho_k = \frac{1}{DI + 1} \quad (3)$$

Ultimately, the damage intensity for each actuator-sensor path is calculated and aggregated up to reconstruct the probabilistic damage images, wherein locations exhibiting a high probability of damage are highlighted in color.

Figure 12 illustrates the outcomes of the RAPID algorithm for damage scenario D3, damage size DS2 under boundary condition BC2, and working frequency of $F2 = 220$ kHz, employing the three most effective DIs: cross-correlation maximum percentage difference (CCMPD), area under frequency spectrum (FAUC), and variance (VAR). The location of the artificial damage is indicated by a blue circle. The probabilistic visualizations for the reported damage indices demonstrate that the dark red areas are indicative of a high likelihood of defect occurrence. In contrast, Fig. 13 depicts the probabilistic imaging results for the damage scenario D4, in which two distinct artificial damages (DS1 and DS2) were positioned on opposing halves of the pressure vessel while it was subjected to boundary condition BC1. The excitation of GW signals was conducted at a frequency of $F2 = 220$ kHz. It can be observed that when multiple damage locations are present, their positions can be identified, although the contrast with the background is diminished.

4.3 Quantitative Evaluation Metric for Damage Visualization. The efficacy of the damage visualization has been evaluated using the prior knowledge of the location of artificial damages and the sensitivity-specificity relationship as outlined in Ref. [33]. The specificity and sensitivity of damage localization and detection were determined for each visualized damage, based on the true known location of the damage. The diagnostic images created with different DIs were evaluated by histogram binarization, with 300 threshold points employed for this purpose. Subsequently, receiver operating characteristic curves (ROC) were generated by plotting the true positive ratio against the false positive ratio. The ROC curves for the DIs highlighted in Figs. 14 and 15 are shown in Figs. 12 and 13, respectively. These curves are exemplary and demonstrate high diagnostic reliability. In order to obtain a quantitative value for the performance of defect detection and localization, the area under ROC curve (AUC-ROC) was considered as a measure of performance for each DI and artificial defect case. The results of the evaluation for all 15 DIs in relation to eight different artificial damage variations are displayed in Fig. 16. The variations in excitation frequency and boundary conditions are indicated by different shapes. When comparing the performance of all DIs under varying experimental conditions, it is evident that there is an almost constant value (80%) of damage detectability and localisability for CCMPD, FAUC, and VAR, which is independent of the current test conditions.

5 Conclusion

A fundamental aspect of the design of a sensor network for damage detection and localization within the COPV was the comprehension of the characteristics of GWs. The data revealed the presence of multiple wave modes, with two dominant modes exhibiting different speeds. A significant discovery was the constant phase velocity, which remained unaffected by direction. This led to the assumption that orthotropy has a reduced impact on wave propagation, which in turn simplifies data processing for damage identification. The sensitivity and attenuation behavior of these modes were subjected to close examination. Ultimately, a sensor network was constructed to provide full coverage of the structure under investigation, based on the previously acquired knowledge regarding wave propagation within the COPV.

The robustness of the network was evaluated through the introduction of artificial damages created by the gluing of weight blocks, which resulted in localized changes in stiffness on the surface of the COPV. A total of eight distinct case scenarios were subjected to measurement in order to obtain a comprehensive multi-layered test data set, which varied factors such as damage size, location, boundary conditions, and excitation frequency. The data were analyzed using the RAPID approach, which employed 15 different DIs and both ultrasonic and time-series features. Overall, the sensor network demonstrated the capacity to effectively detect and localize damage, although the sensitivity of the DIs exhibited variability contingent on the specific experimental conditions. In order to obtain a quantitative metric regarding the effectiveness of the detection performance, the AUC-ROC was considered for each DI. This resulted in the conclusion that artificial damages are most effectively detected using indices such as VAR, CCMPD, or FAUC.

It is possible that real defects, such as delamination or impact damage, may display different characteristics with regard to damage detection and GW feature relationships. In order to facilitate the selection process for DI, it would be beneficial to employ additional quantitative evaluation metrics, such as the AUC-ROC.

Acknowledgment

The authors acknowledge funding from the German Ministry of Economic Affairs and Climate Actions within the QI-Digital initiative.

Conflict of Interest

There are no conflicts of interest.

Data Availability Statement

The datasets generated and supporting the findings of this article are obtainable from the corresponding author upon reasonable request.

References

- [1] Munzke, D., Duffner, E., Eisermann, R., Schukar, M., Schoppa, A., Szczepaniak, M., Strohacker, J., and Mair, G. W., 2020, "Monitoring of Type IV Composite Pressure Vessels With Multilayer Fully Integrated Optical Fiber Based Distributed Strain Sensing," *Mater. Today: Proc.*, **34**(1), pp. 217–223.
- [2] Jones, H. W., 2017, "Oxygen Storage Tanks are Feasible for Mars Transit," International Conference on Environmental Systems, Paper No. ICES-2017-89.
- [3] Memmolo, V., Maio, L., and Ricci, F., 2022, "Assessment of Damage in Composite Pressure Vessels Using Guided Waves," *Sensors*, **22**(14), p. 5182.
- [4] McLaughlan, P. B., Forth, S. C., and Grimes-Ledesma, L., 2011, "Composite Overwrapped Pressure Vessels", A Primer, NASA Johnson Space Center, Houston, TX, Special Publication NASA/SP-2011-572.
- [5] Greene, N., Saulsberry, R., Leifeste, M. R., Yoder, T., Keddy, C. P., Forth, S. C., and Russell, R. W., 2010, "Composite Overwrapped Pressure Vessel (COPV) Stress Rupture Testing," 4th IAASS Conference - Making Safety Matter, SP-680.
- [6] EN ISO 11623:2023, 2023, *Gas Cylinders - Composite Cylinders and Tubes - Periodic Inspection and Testing*, International Organization for Standardization (ISO), Geneva, Switzerland.

- [7] Dahmene, F., Yaacoubi, S., Mountassir, M. E., Langlois, C., and Bardoux, O., 2016, "Towards Efficient Acoustic Emission Testing of COPV, Without Felicity Ratio Criterion, During Hydrogen-Filling," *Int. J. Hydrogen Energy*, **41**(2), pp. 1359–1368.
- [8] David-West, O., Montalvão, D., and Haritos, G., 2017, "A Review of Structural Health Monitoring Techniques as Applied to Composite Structures," *Struct. Durab. Health Monit.*, **11**(2), pp. 91–147.
- [9] Mueller, I., Memmolo, V., Tschöke, K., Moix-Bonet, M., Möllenhoff, K., Golub, M. V., Venkat, R. S., Lugovtsova, Y., Eremin, A., and Moll, J., 2022, "Performance Assessment for a Guided Wave-Based SHM System Applied to a Stiffened Composite Structure," *Sensors*, **22**(19), p. 7529.
- [10] Rose, J. L., Ditri, J. J., Pilarski, A., Zhang, J., Carr, F., and McNight, W., 1992, "A Guided Wave Inspection Technique for Nuclear Steam Generator Tubing," *Non-Destruct. Test.*, **27**(6), pp. 307–310.
- [11] Alleyne, D. N., and Cawley, P., 1996, "The Excitation of Lamb Waves in Pipes Using Dry-Coupled Piezoelectric Transducers," *ASME J. Nondestruct. Eval.*, **15**(1), pp. 11–20.
- [12] Demma, A., Cawley, P., Lowe, M. J. S., and Roosenbrand, A. G., 2003, "The Reflection of the Fundamental Torsional Mode From Cracks and Notches in Pipes," *J. Acoust. Soc. Am.*, **114**(2), pp. 611–625.
- [13] Da, Y., Wang, B., Liu, D. Z., and Qian, Z., 2020, "An Analytical Approach to Reconstruction of Axisymmetric Defects in Pipelines Using T(0,1) Guided Waves," *Appl. Math. Mech.*, **41**(10), pp. 1479–1492.
- [14] Ismail, Z., Mustapha, S., and Tarhini, H., 2021, "Optimizing the Placement of Piezoelectric Wafers on Closed Sections Using a Genetic Algorithm—Towards Application in Structural Health Monitoring," *Ultrasonics*, **116**(8), p. 106523.
- [15] Li, F., Li, F., Liu, Z., Liu, Z., Sun, X., Sun, X., Li, H., Li, H., Meng, G., and Meng, G., 2015, "Propagation of Guided Waves in Pressure Vessel," *Wave Motion*, **52**(1), pp. 216–228.
- [16] Yang, B., Xiang, Y., Xuan, F.-Z., Hu, C., Xiao, B., Zhou, S., and Luo, C., 2019, "Damage Localization in Hydrogen Storage Vessel by Guided Waves Based on a Real-Time Monitoring System," *Int. J. Hydrogen Energy*, **44**(40), pp. 22740–22751.
- [17] Li, Q., Luo, Z., Hu, G., and Zhou, S., 2022, "A New Probabilistic Ellipse Imaging Method Based on Adaptive Signal Truncation for Ultrasonic Guided Wave Defect Localization on Pressure Vessels," *Sensors*, **22**(4), p. 1540.
- [18] Lugovtsova, Y., and Prager, J., 2018, "Structural Health Monitoring of Composite Pressure Vessels Using Guided Ultrasonic Waves," *Insight*, **60**(3), pp. 139–44.
- [19] Lugovtsova, Y., Bulling, J., Boller, C., and Prager, J., 2019, "Analysis of Guided Wave Propagation in a Multi-layered Structure in View of Structural Health Monitoring," *Appl. Sci.*, **9**(21), p. 4600.
- [20] Su, Z., and Ye, L., 2009, *Identification of Damage Using Lamb Waves: From Fundamentals to Applications*, 1st ed., Springer-Verlag, London.
- [21] Gazis, D. C., 1959, "Three-Dimensional Investigation of the Propagation of Waves in Hollow Circular Cylinders. II. Numerical Results," *J. Acoust. Soc. Am.*, **31**(5), pp. 573–578.
- [22] Demma, A., Cawley, P., Lowe, M. J. S., Roosenbrand, A. G., and Pavlakovic, B., 2004, "The Reflection of Guided Waves From Notches in Pipes: A Guide for Interpreting Corrosion Measurements," *NDT & E Int.*, **37**(3), pp. 167–180.
- [23] Fong, K. L. J., 2005, "A Study of Curvature Effects on Guided Elastic Waves," Ph.D. dissertation, Imperial College London, London. <https://www.imperial.ac.uk/media/imperial-college/research-centres-and-groups/non-destructive-evaluation/Fong-K-L-J.PDF>.
- [24] Meitzler, A., 1961, "Mode Coupling Occurring in the Propagation of Elastic Pulses in Wires," *J. Acoust. Soc. Am.*, **33**(4), pp. 435–445.
- [25] Verma, B., Mishra, T. K., Balasubramaniam, K., and Rajagopal, P., 2014, "Interaction of Low-Frequency Axisymmetric Ultrasonic Guided Waves With Bends in Pipes of Arbitrary Bend Angle and General Bend Radius," *Ultrasonics*, **54**(3), pp. 801–808.
- [26] Giurgiutiu, V., 2014, *Structural Health Monitoring With Piezoelectric Wafer Active Sensors*, Academic Press, London.
- [27] Kiefer, D. A., 2024, GEWtool (v1.5). Zenodo, doi: 10.5281/zenodo.12734260.
- [28] Hu, C. J., Yang, B., Wu, Y., Xiao, B., and Xuan, F., 2019, "Guided Wave Propagating Behavior in Composite Pressure Vessel," 7th Asia-Pacific Workshop on Structural Health Monitoring, Hong Kong, China, Nov. 12–15.
- [29] Kudela, P., Radziński, M., Moix-Bonet, M., Willberg, C., Lugovtsova, Y., Bulling, J., Tschöke, K., and Moll, J., 2022, "Dataset on Full Ultrasonic Guided Wavefield Measurements of a CFRP Plate With Fully Bonded and Partially Debonded Omega Stringer," *Data Brief*, **42**(3), p. 108078.
- [30] Huang, C.-F., and Tseng, Y., 2005, "The Coverage Problem in a Wireless Sensor Network," *Mobile Netw. Appl.*, **10**(4), pp. 519–528.
- [31] Velsor, J. K. V., Gao, H., and Rose, J. L., 2007, "Guided-Wave Tomographic Imaging of Defects in Pipe Using a Probabilistic Reconstruction Algorithm," *Insight*, **49**(9), pp. 532–537.
- [32] Loendersloot, R., Buethe, I., Michaelides, P., Bonet, M. M., and Lampeas, G., 2015, "Damage Identification in Composite Panels—Methodologies and Visualisation," Smart Intelligent Aircraft Structures (SARISTU), Moscow, Russia, May 19–21.
- [33] Yilmaz, B., Ba, A., Jasiūnienė, E., Bui, H.-K., and Berthiau, G., 2020, "Evaluation of Bonding Quality with Advanced Nondestructive Testing (NDT) and Data Fusion," *Sensors*, **20**(18), pp. 5127.